



Department of Electrical and Computer Engineering
COMSATS University Islamabad, Attock Campus

Lab Rubrics Form	
Lab Name	Array Manipulation, While Loop Programming, and Introduction to Software Defined Radio (SDR) using LabVIEW
Lab Number	2
Submitted by:	
Student Names	Registration #
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Rubrics		Marks		
		1	2	3
PLO-5 Modern Tools Usage	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 Communication Skills	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

Lab No: 02

Array Manipulation, While Loop Programming, and Introduction to Software Defined Radio (SDR) using LabVIEW

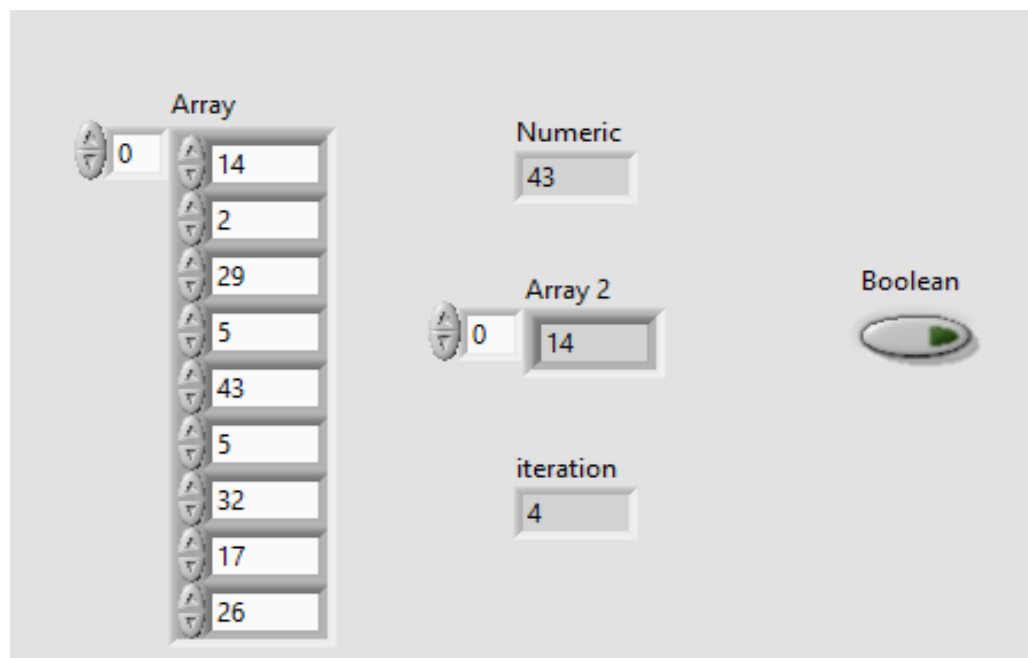
Objective

- To understand the implementation of While Loop in LabVIEW for iterative execution and array processing.
- To develop algorithms for performing array manipulation using built-in array functions such as Array Size and indexing.
- To learn how to search for the first array element greater than a specified value using LabVIEW programming logic.
- To study the basic architecture, features, and applications of the NI USRP-2922 Software Defined Radio (SDR) and its integration with LabVIEW.
- To gain practical knowledge of LabVIEW programming for solving engineering and communication system problems.

Task No: 01

Create a VI in while loop. Create the logic to find out which (the first) index in the array that have the number greater than 30. See front panel below.

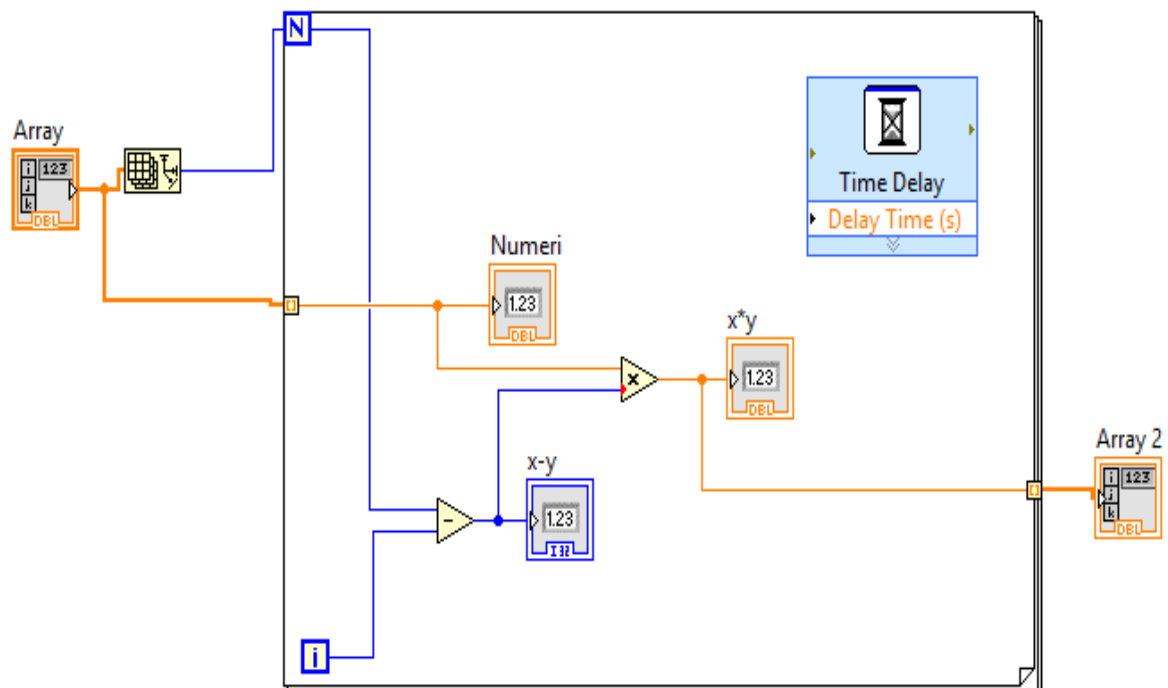
Output:

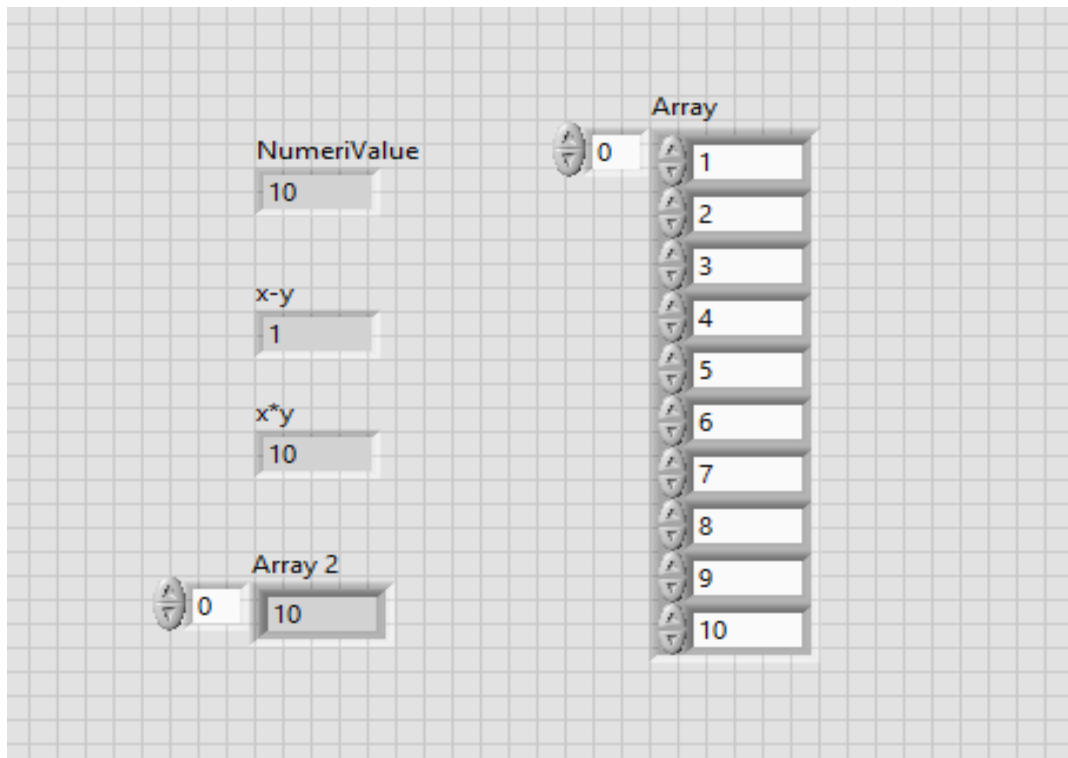


Task No: 02

Make an array $v = [1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10]$, develop an algorithm such that the first element of the vector is multiplied by $\text{length}(v)$, second element by $\text{length}(v)-1$ and similarly the last element i.e. 10 is multiplied by $\text{length}(v)-9$. The final output array should be $f = [10\ 18\ 24\ 28\ 30\ 30\ 28\ 24\ 18\ 10]$. The algorithm devised should only use the array size function of array v to achieve array f .

Ouput:





Task No: 03

Question #: 01

What is the frequency range of USRP kit 2922?

USRP-2922, a product designed by National Instruments, is a USRP Software Defined The USRP Software Defined Radio Device consists of a transmitter and a receiver that support a wide frequency range of 400 MHz to 4.4 GHz.

Frequency Accuracy:

In the USRP-2922 device, both the transmitter and receiver operate at a frequency step less than 1 kHz and frequency accuracy of 2.5 ppm.

Example:

The NI USRP-2922 can effectively be used in WiFi, WiMax, and ISM band communication applications.

Question No: 02

How many ports are available for transmission and reception on USRP kit 2922?

The USRP-2922 has a total of RF ports:

TX1: This is the transmission port.

TX2: This is the 2nd transmission port.

RX1: This is one of the reception ports.

RX2: This is the second reception port¹².

- These ports allow the USRP-2922 to send and receive signals for various communication applications.
- Support for multiple protocols and applications (e.g., MIMO, diversity).
- Flexibility in system design and experimentation.

Question No: 03

What are the hardware components which are performed in computer/LabVIEW SDR?

LabVIEW development system provides an ideal way to interface with NI USRP hardware for the development and exploration of communications algorithms that process received signals and synthesize signals for transmission.

Hardware Components:

1. RF Front End

- **Antennas** (TX/RX) (transmit and receive).
- **Amplifiers** (low-noise, power).
- **Filters** (band-pass, band-stop).
- **Switches** (TX/RX, frequency).

2. Analog-to-Digital Converter (ADC)

- Sampling rates: 100 MS/s to 1 GS/s

3. Digital-to-Analog Converter (DAC)

- Sampling rates: 100 MS/s to 1 GS/s

4. Field-Programmable Gate Array (FPGA)

- Signal processing, decoding, encoding.

LabVIEW SDR Specific Components:

- 1.** National Instruments (NI) USRP.
- 2.** NI FlexRIO.
- 3.** NI PXIe (Peripheral Component Interconnect Express).

Conclusion

In this lab, the While Loop was successfully implemented to process arrays and determine the first index containing a value greater than 30. An algorithm was also developed using only the Array Size function to generate the required output array, demonstrating efficient array manipulation techniques in LabVIEW. Furthermore, the fundamental concepts of Software Defined Radio (SDR) were explored by studying the NI USRP-2922, including its operating frequency range, RF ports, and major hardware components. Overall, the lab enhanced understanding of LabVIEW programming, array operations, iterative structures, and the practical use of SDR technology in modern wireless communication systems.